

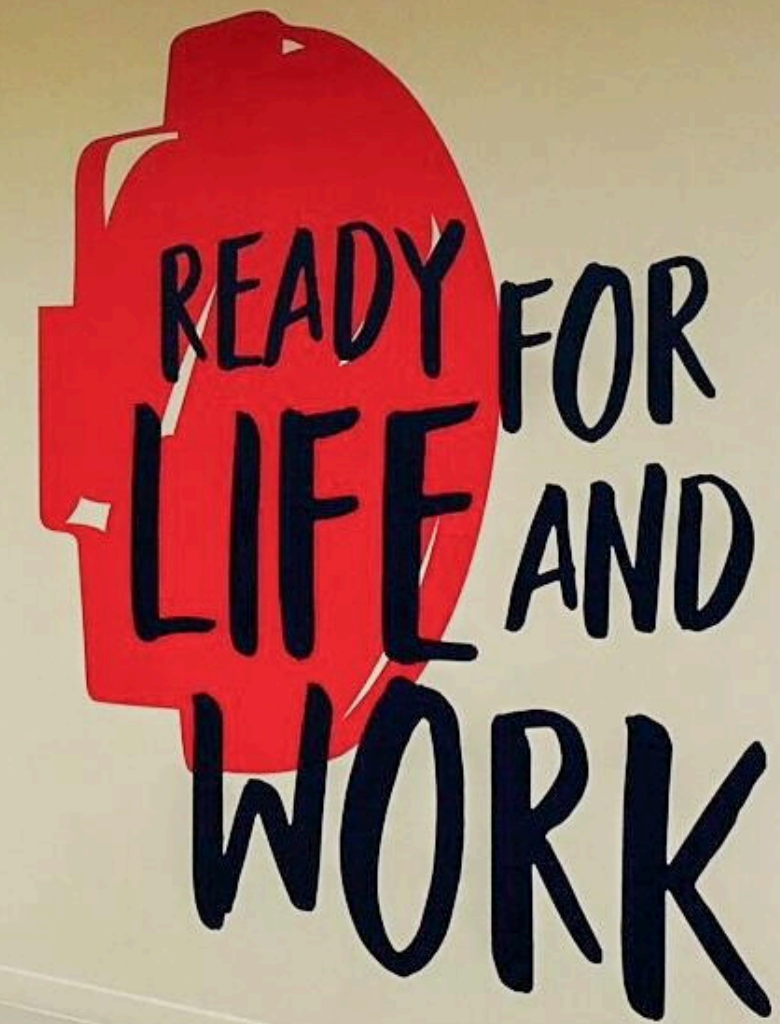
Foundations of Machine Learning

COSC2753/COSC2812

Machine Learning

Semester 2, 2026

Dr. Bao Nguyen



Agenda



- Course Overview
 - Course Objectives
 - Syllabus and others
- What is Machine Learning?
 - Applications
 - Problem types

Course Overview

Teaching team



Dr. Nguyen Thien Bao

Lecturer and Course Coordinator

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Office: 2.4.27, SGS Campus

A boring bit about your lecturer



- **Dr. Bao Nguyen**

- Ph.D. Computer Science @ University of Trento, Italy
- Visiting scholar, Surgical Planning Laboratory (SPL), Harvard, MA, USA
- Researcher, NeuroInformatics Lab (NILab), FBK, Trento, Italy
- Lecturer, SSET, RMIT, Vietnam

- **Research interest:**

- Computer Vision, Image Processing, Medical Imaging
- Machine Learning, Deep Learning, Fashion Learning
- Digital Twin, Smart Agriculture
- ...

Course objectives



Learn about machine learning (ML) conceptually and practically

- Introduce basic machine learning concepts
- How to **apply** ML techniques to a range of problems, using open-source ML toolkits
- How to **analyze**:
 - What the algorithms are doing
 - Data sets & outputs from the applications
- Some **terminology** and **concepts** of ML approaches may not be covered in the course

Course information



Desirable

- Knowledge of Python
- Calculus (gradients)
- Linear algebra
- Optimisation
- Probability and statistics
- Algorithms

If you are not strongly confident in any of these areas then no worries, they just help you to understand the materials more easily

Course workload & assessments



Contact hours (every week):

- Lectures (1.5 hours), Tutorials (1.5 hours)

Assessment:

- Intro to Machine Learning – Individual (20%)
- Machine Learning Project – Group (40%)
- Presentation & Interview – Individual (40%)

Group work



- You will **form your own group** for assessment in this course (**week 2-3**)
- Task allocation must be equally distributed and fair
- Please raise any potential or occurring issues to the Course Coordinator as soon as possible
- Provide **at least two weeks before the due date** to resolve group issues

Assessment policies



- I will endeavor to finish grading within 14 business days after the due date
- Feedback will be provided via Canvas

Communication



Discussion Board on Canvas for Q&As, discussions etc.

- https://rmit.instructure.com/courses/171525/discussion_topics
- Please don't post solutions there (or anywhere)

Consultation hours

- Please confirm a calendar booking before coming
- No consultation during lunch time please
- Without appointment I may not be in the office or busy doing something else

Communication



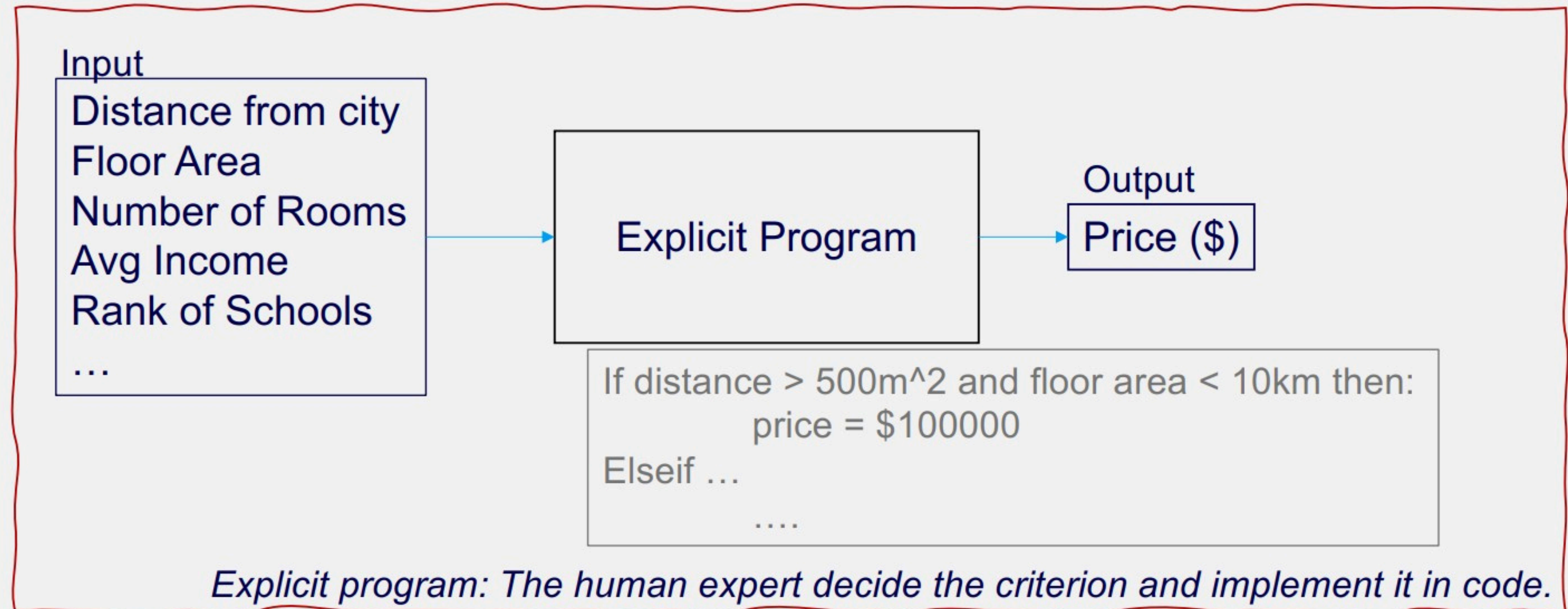
Email works best

- Please try to mention COSC2753/COSC2812 or ML in the subject line
- I will reply ASAP within 02 business days

Make sure to read the Course Announcements

Foundations of Machine Learning

What is Machine Learning?

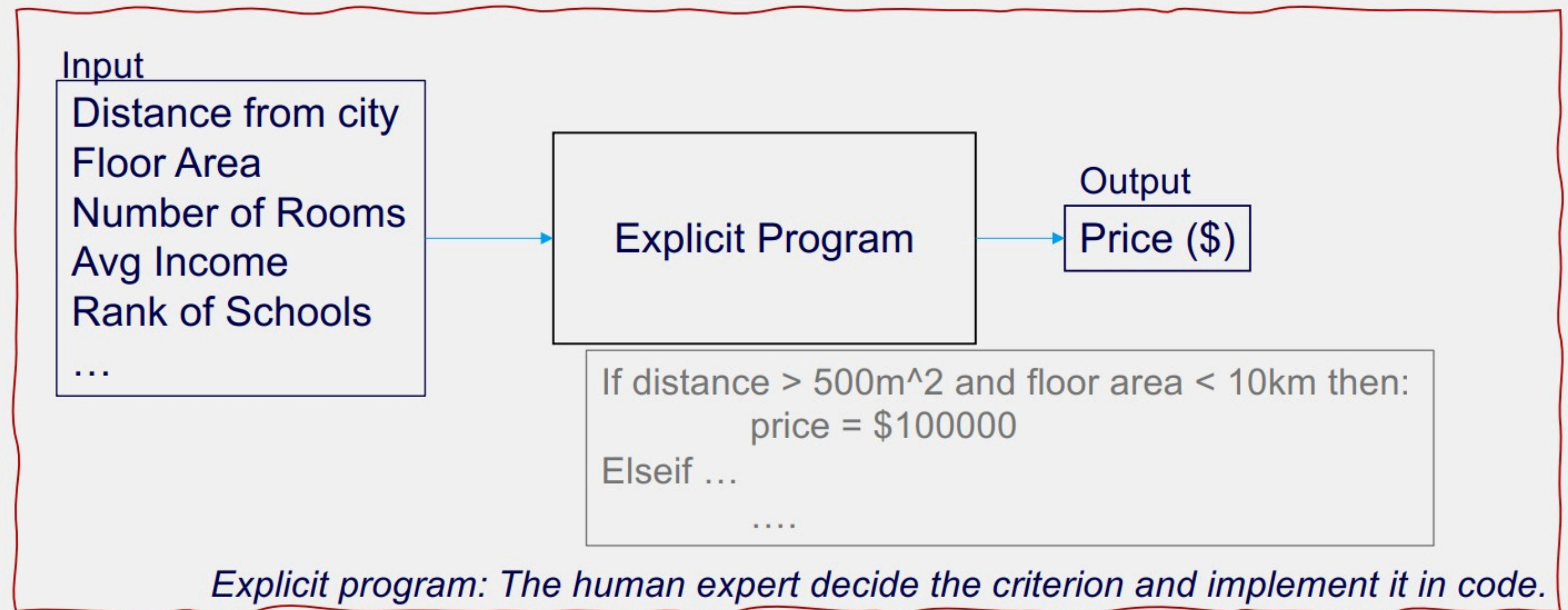


What is Machine Learning?

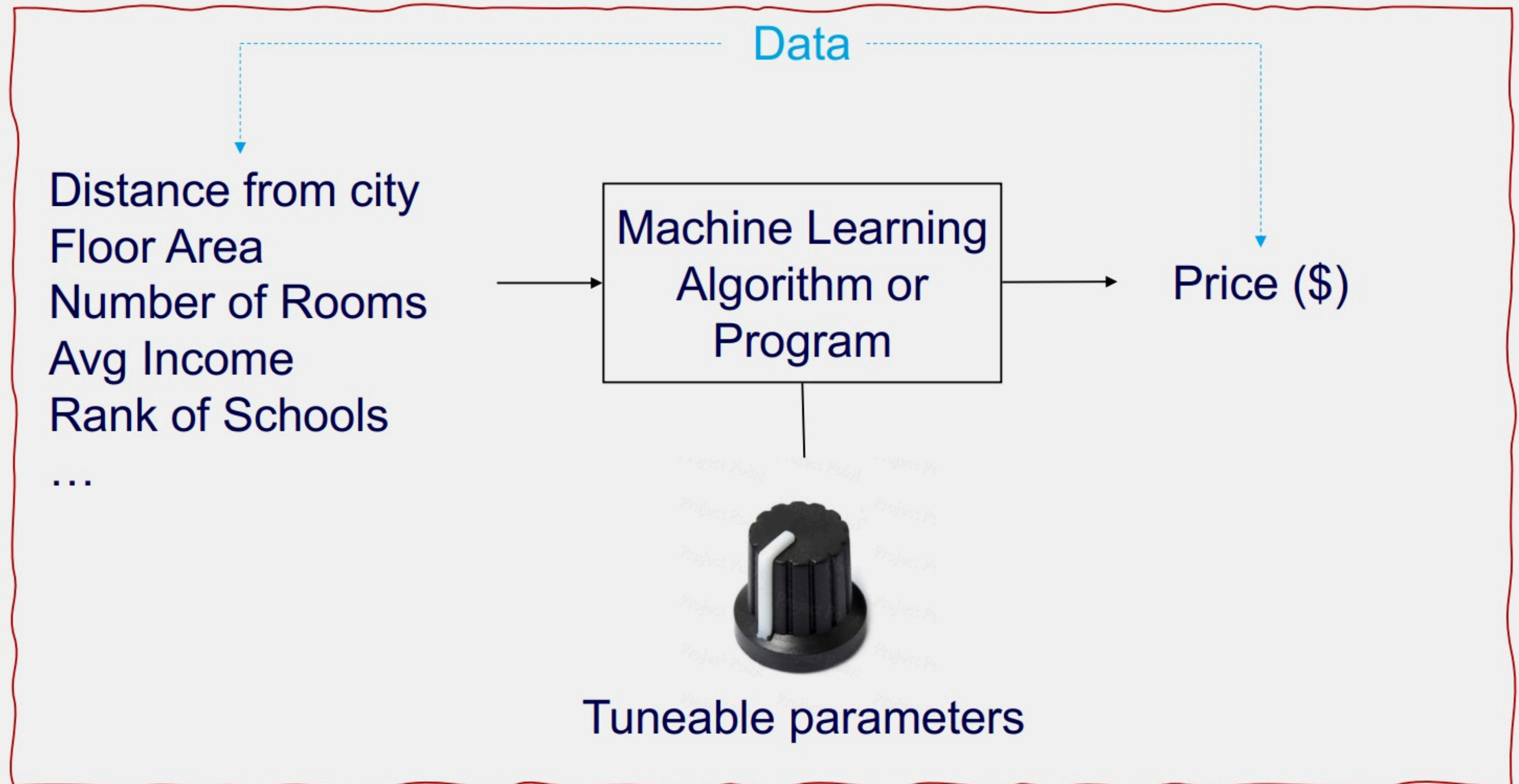


“Machine learning is the field of study that gives the **ability to learn without being explicitly programmed.**”

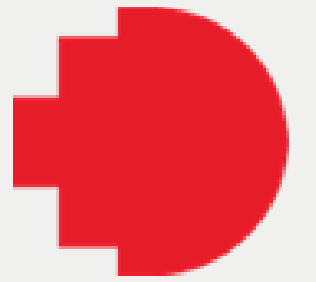
Arthur Samuel (1959)



House price prediction example



What is Machine Learning?



Tom Mitchell

More technically:

“A computer program is said to learn:

- Some class of **tasks T**
- From **experience E**, and
- **Performance measure P**

If its performance at tasks in T, as measured by P, improves with experience E”

Tom Mitchell (1998)

Task: unknown target function



The **Task** can be expressed as an **unknown target function**:

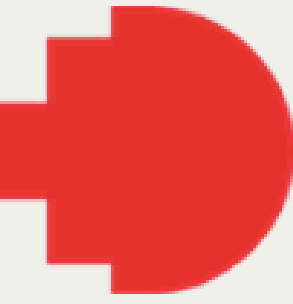
$$y = f(\mathbf{x})$$

- Attributes of the task: $\mathbf{x}$
- Unknown function: $f(x)$
- Output of the function: y

ML finds a **Hypothesis**, h , which is a function (or model) to approximate the unknown target function

$$h(\mathbf{x}) \approx f(\mathbf{x})$$

Experience



The **Experience** is typically a data set, D , of values:

$$D = \left\{ \left(\mathbf{x}^{(i)}, f(\mathbf{x}^{(i)}) \right) \right\}_{i=1}^n$$

Attributes of task: $\mathbf{x}^{(i)}$

Output of the Unknown function: $f(\mathbf{x}^{(i)})$

A data instance (or data point) i is a tuple (*assume supervised learning*): $\left(\mathbf{x}^{(i)}, f(\mathbf{x}^{(i)}) \right)$ (input-output pairs) from observations

Performance



What does success look like?

We would like to measure: $h^*(\mathbf{x}) \approx f(\mathbf{x})$

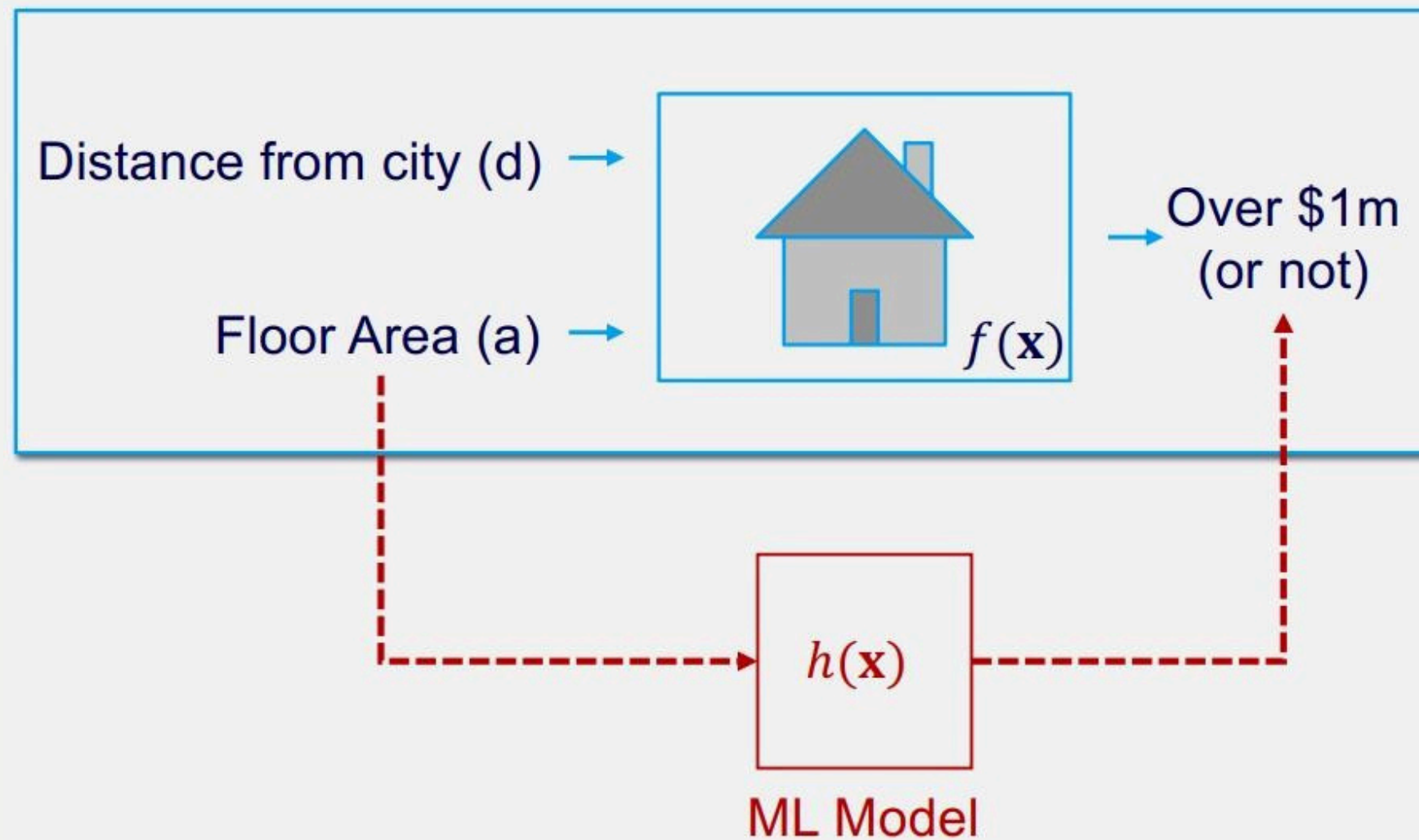
Performance:

- Determines how well the hypothesis matches the experience.
- Note, the performance is measured against the experience, NOT the unknown target function!

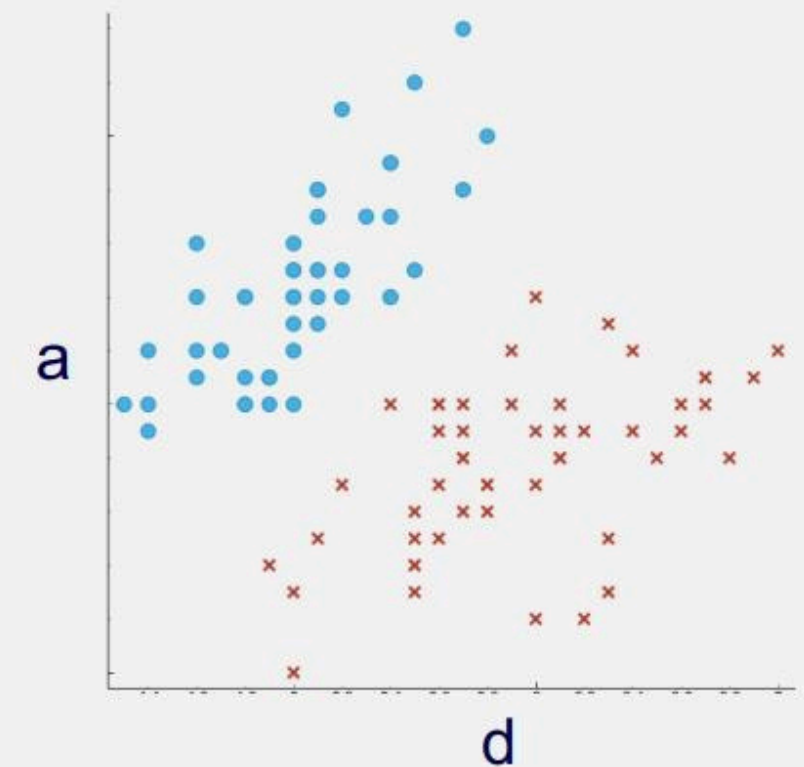
Simple example

House price prediction

Example: House price prediction

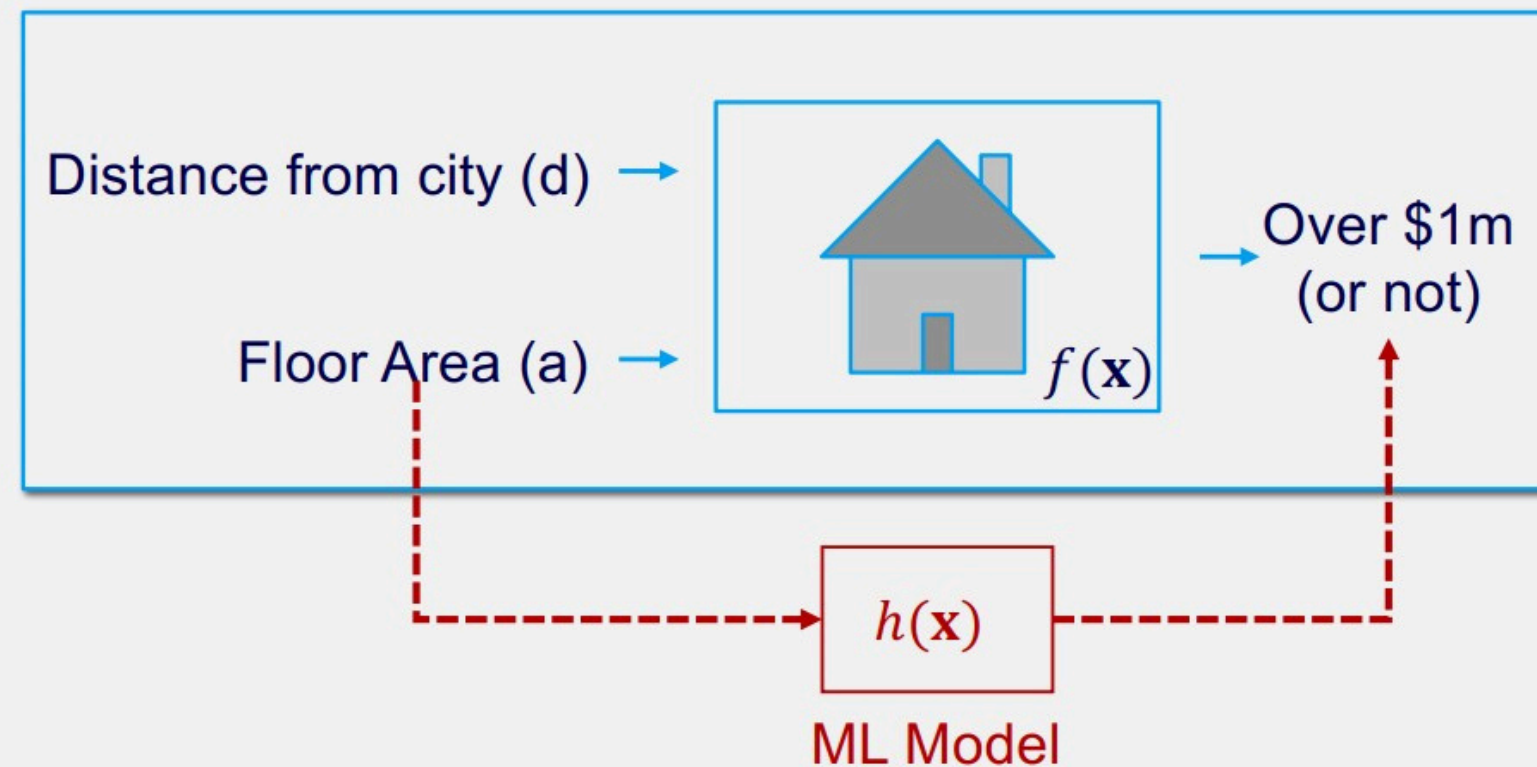
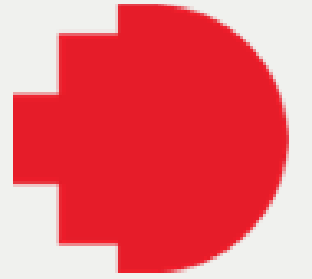


i	d	a	$f(\mathbf{x})$
1	25	150	N
2	10	100	Y
...	...	...	...
n	32	450	Y



What form should $h(\mathbf{x})$ take?

Example: House price prediction



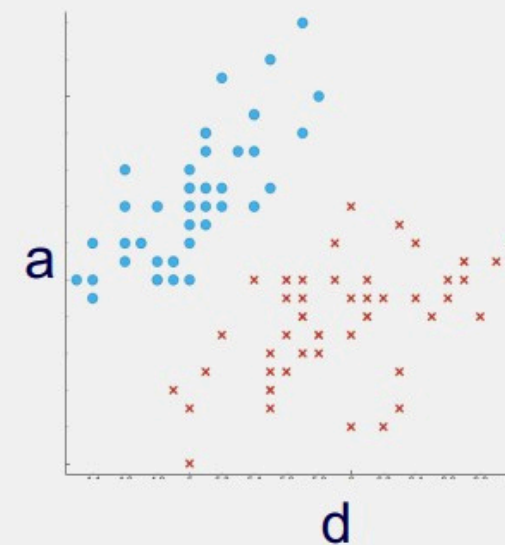
We need a **performance measure** that quantify what “**best**” means.

Let's use the count of how many errors we make**.

Now we combine data and hypothesis space and, select the hypothesis that makes the least number of errors as our **optimal hypothesis** $h^*(\mathbf{x})$.

This is done automatically through the **optimization procedure**.

Once we have the $h^*(\mathbf{x})$
We can use it to predict the
target value for future data
points.



** May not be the best measure. We will learn better ones later in the course.

Types of ML

Types of machine learning problems



- Supervised Learning
 - Classification
 - Regression
- Unsupervised Learning
- Semi-Supervised Learning

Supervised learning



In **supervised learning**, the output is known: $y = f(\mathbf{x})$

Experience: Examples of input-output pairs

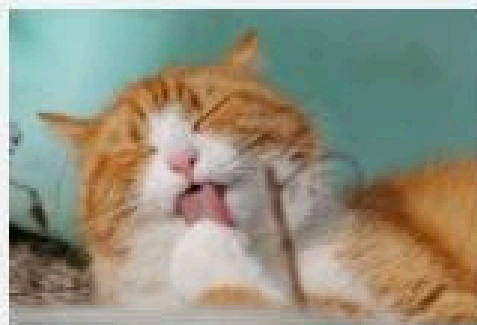
Task:

- Learns a model that maps input to desired output
- Predict the output for new “unseen” inputs.

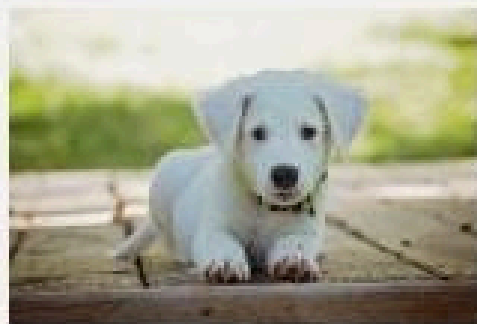
Performance: Error measure of how closely the hypothesis predicts the target output

Two main types of supervised learning: classification & regression

Supervised learning: Examples

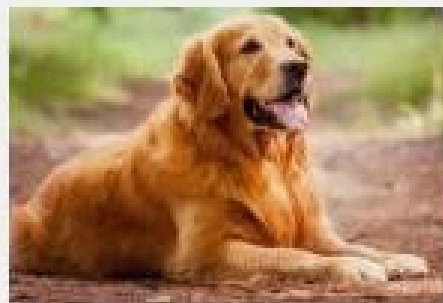


cat



Classifier
(ML Algorithm)

dog

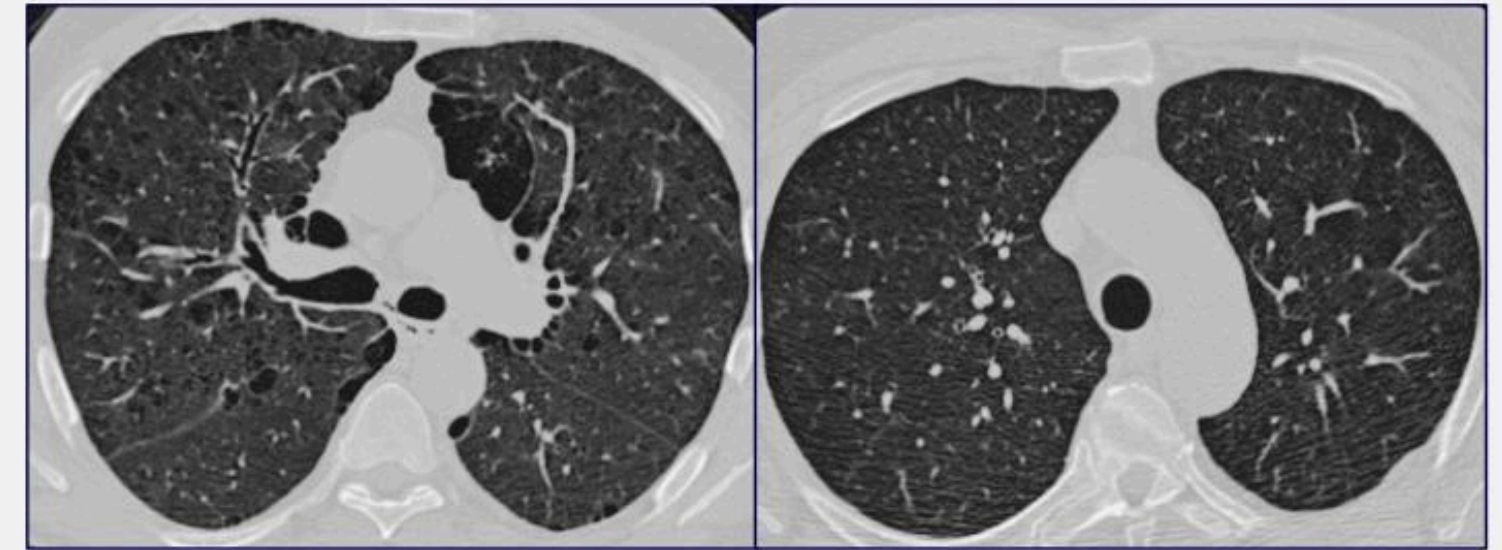


dog

Supervised learning: Examples



Toshiba Advanced Driver Assistance Systems



Axial slices of two 3D-CT images with (left) and without (right) emphysema.

Tennakoon, R., et. al. "Classification of Volumetric Images Using Multi-Instance Learning and Extreme Value Theorem". *IEEE Transactions on Medical Imaging*, 2019.

Unsupervised learning



In **unsupervised learning**, the output is known:

$$? = f(\mathbf{x})$$

Experience: Data set with values for some or all attributes

Task:

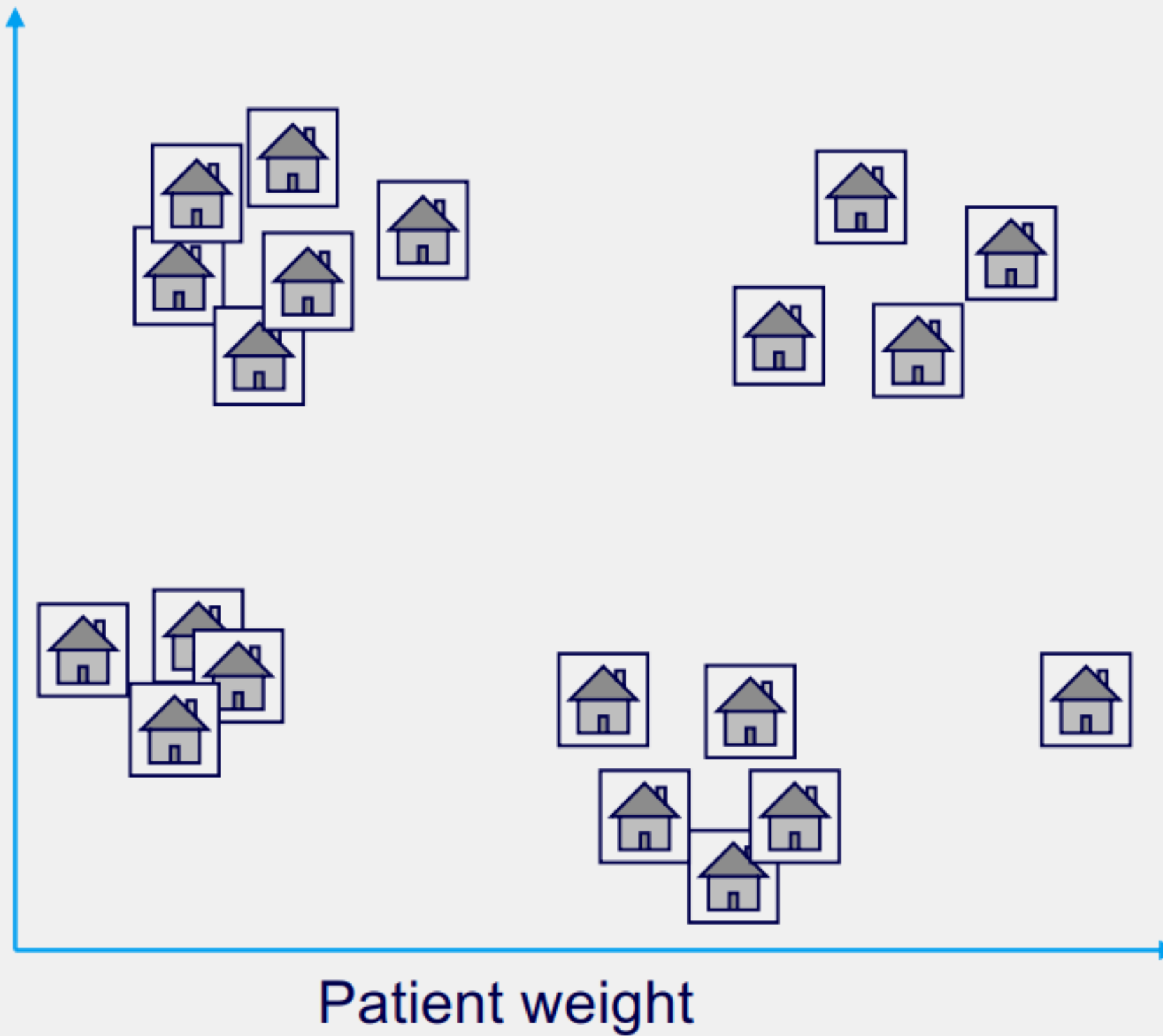
- “Invent” a suitable output
- Identify trends and patterns between the data points

Performance: How well the “invented” output matches the data set

Unsupervised learning: simple clustering



Patient blood
glucose level



Unsupervised learning: computer vision



Smart paint brush with Generative Adversarial Network (GAN).

Types of machine learning problems



Others

- Semi-supervised
- Active learning
- Transfer learning
- Federated learning
- ...

Model complexity

Which hypothesis should be chosen?

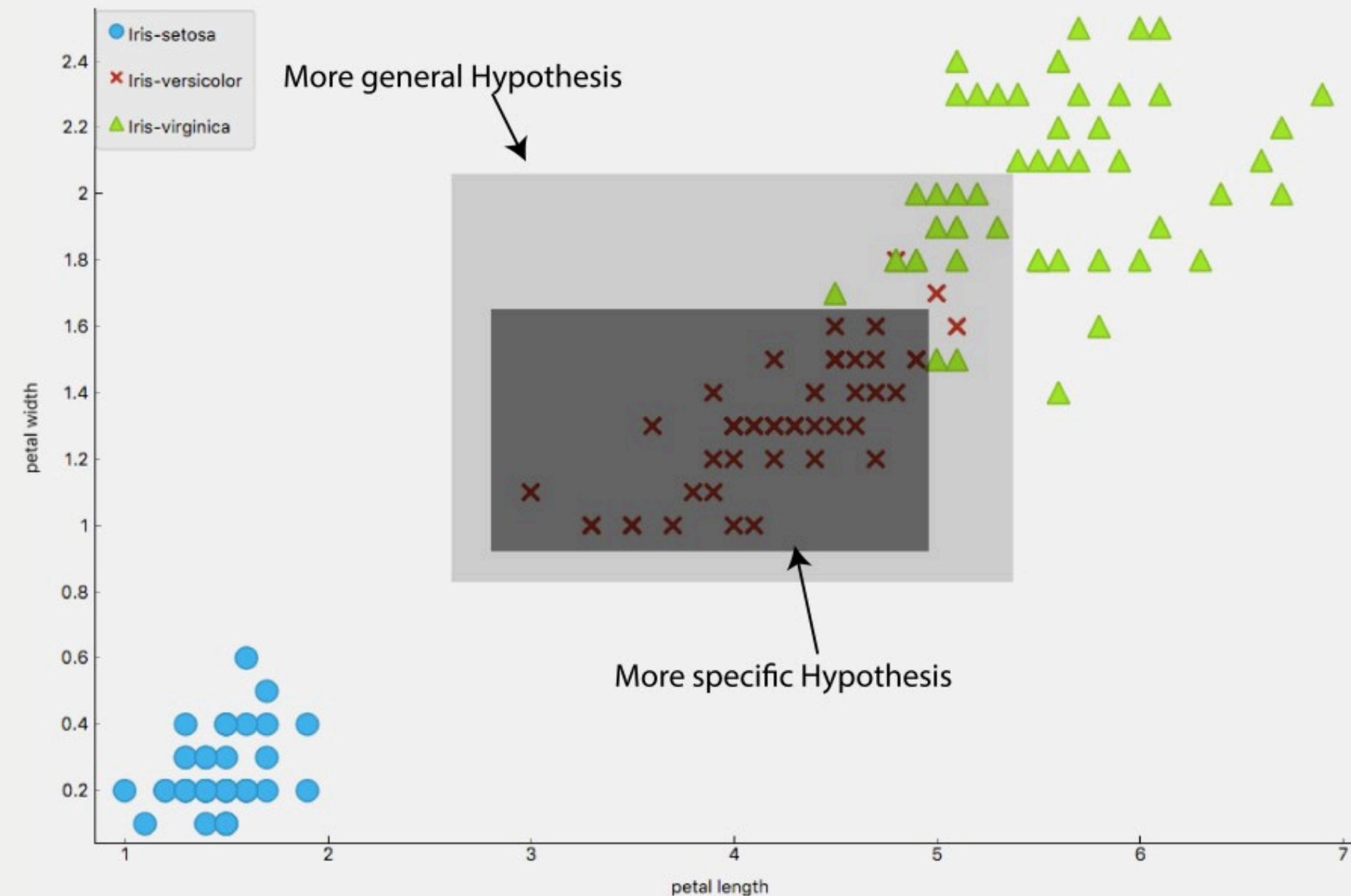
Hypothesis space



The **hypothesis space** is the set of all hypotheses over the state space of a problem that an algorithm is capable of learning

The great ML question:

Which hypothesis should be learnt?



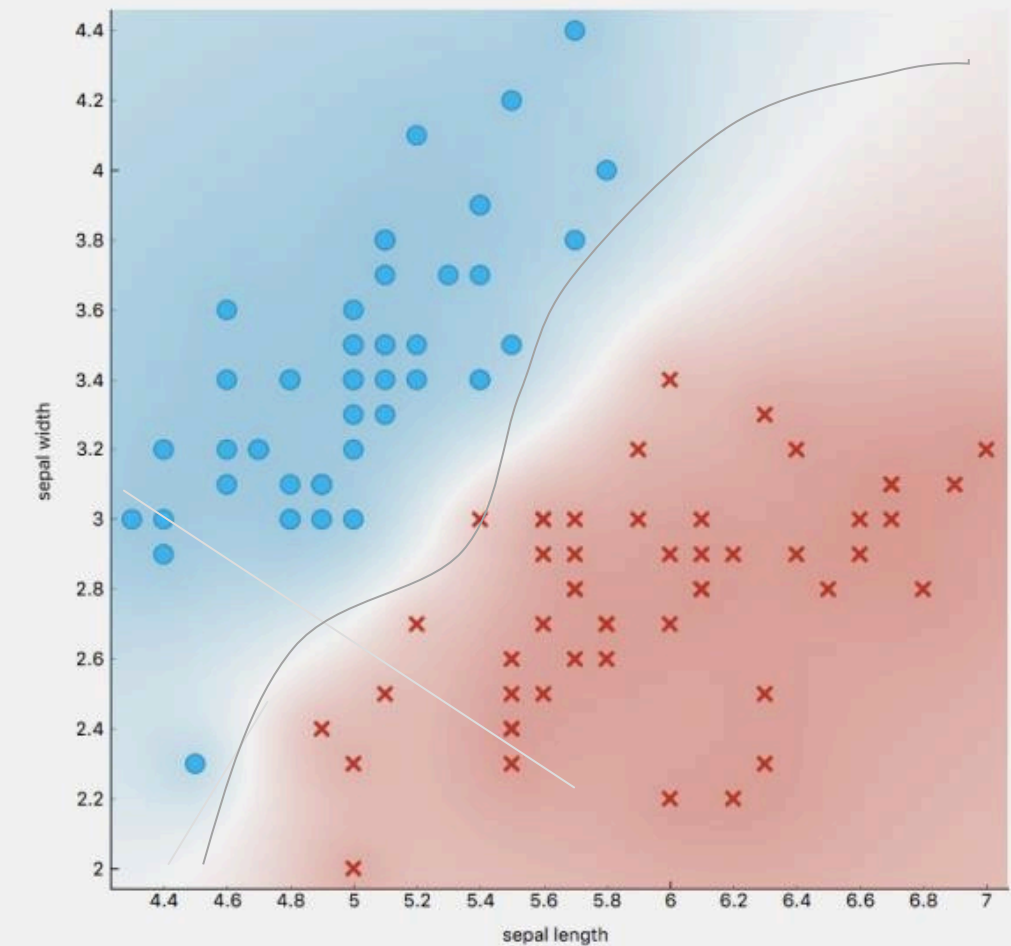
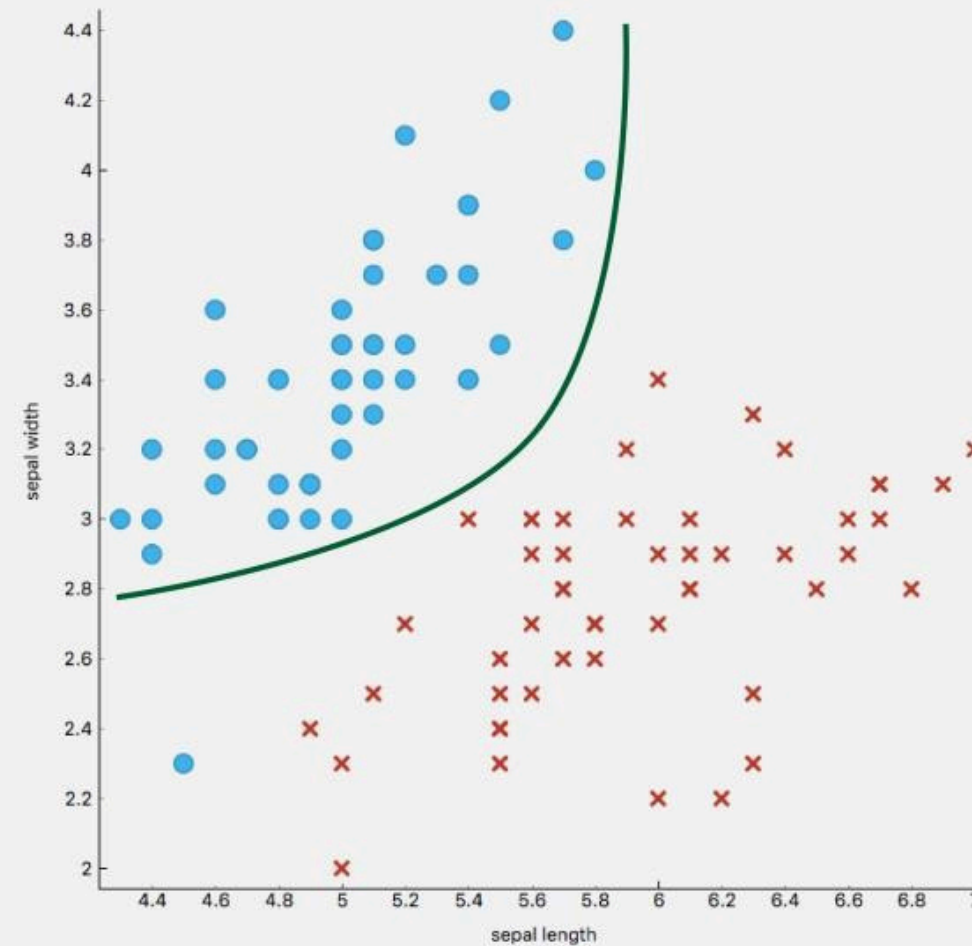
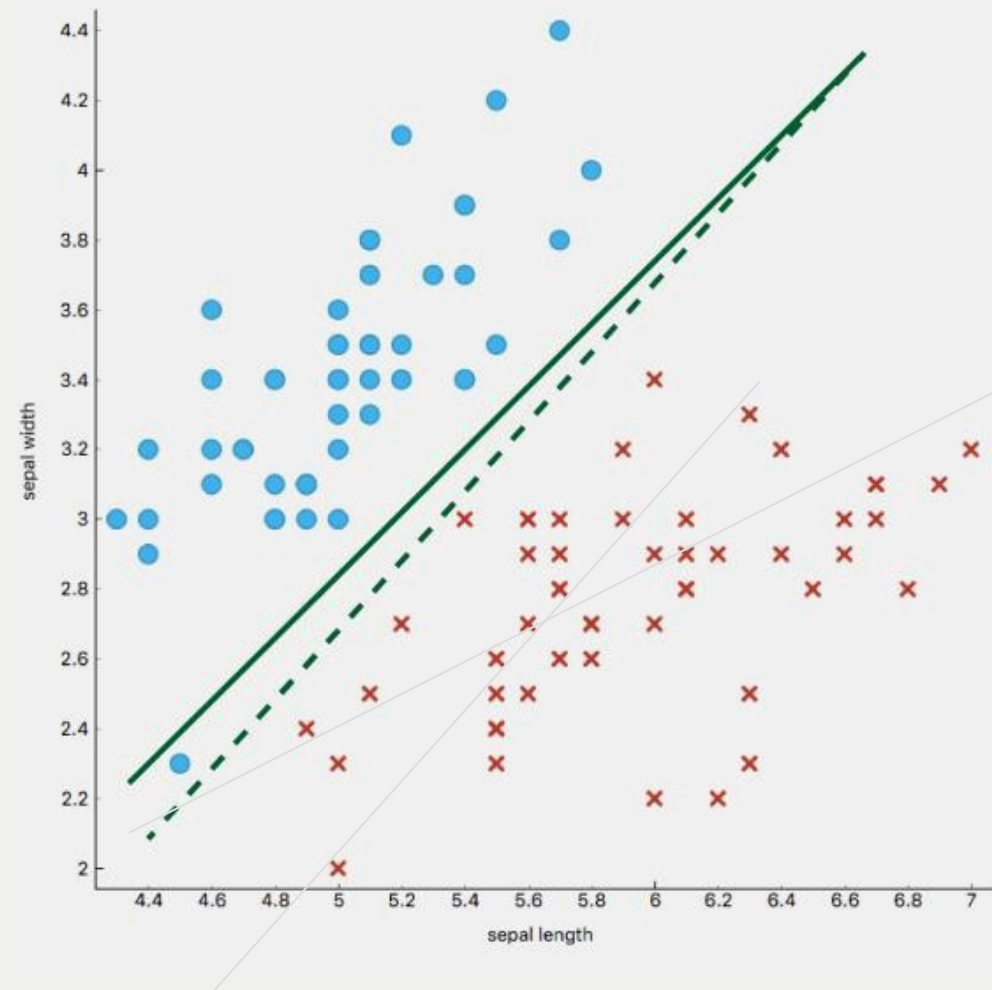
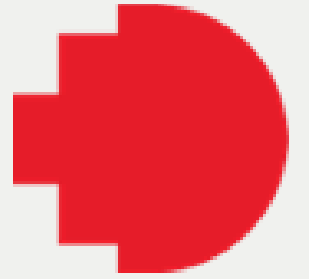
In this example we want to separate (red x) from the rest. Our hypothesis space is all the possible rectangles.

What is the best model?

What model should be learnt?

$$y = ax + b$$

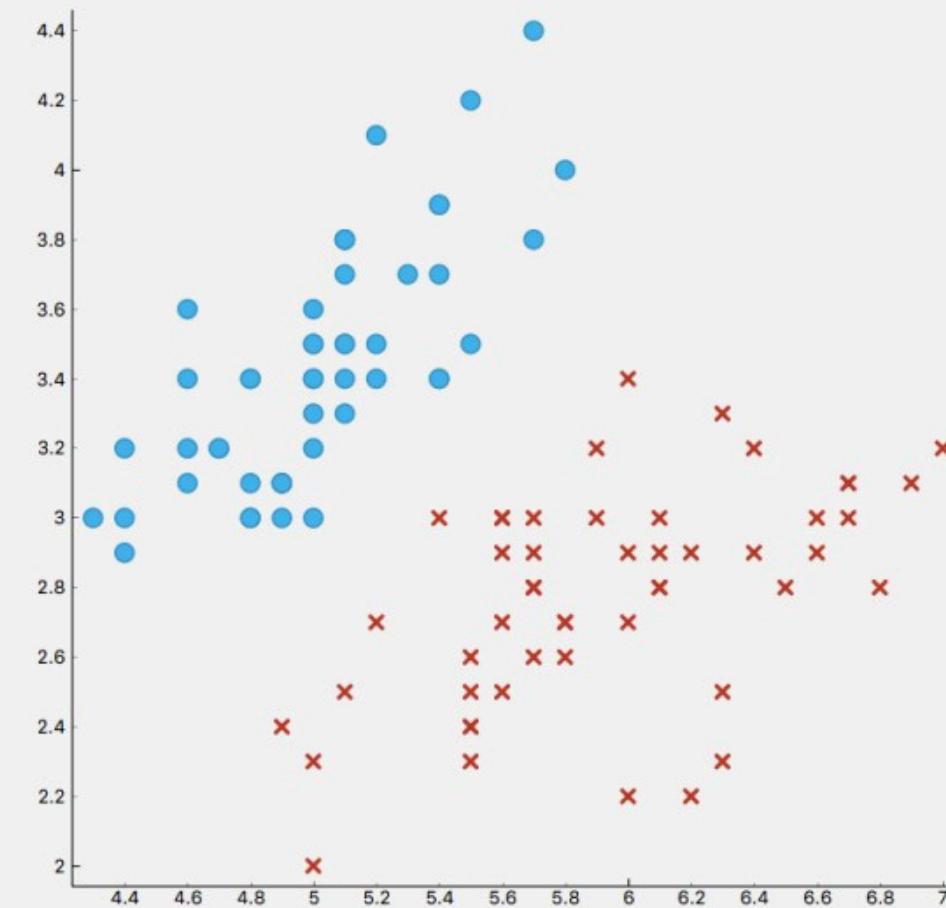
$$y = ax^2 + bx + c$$



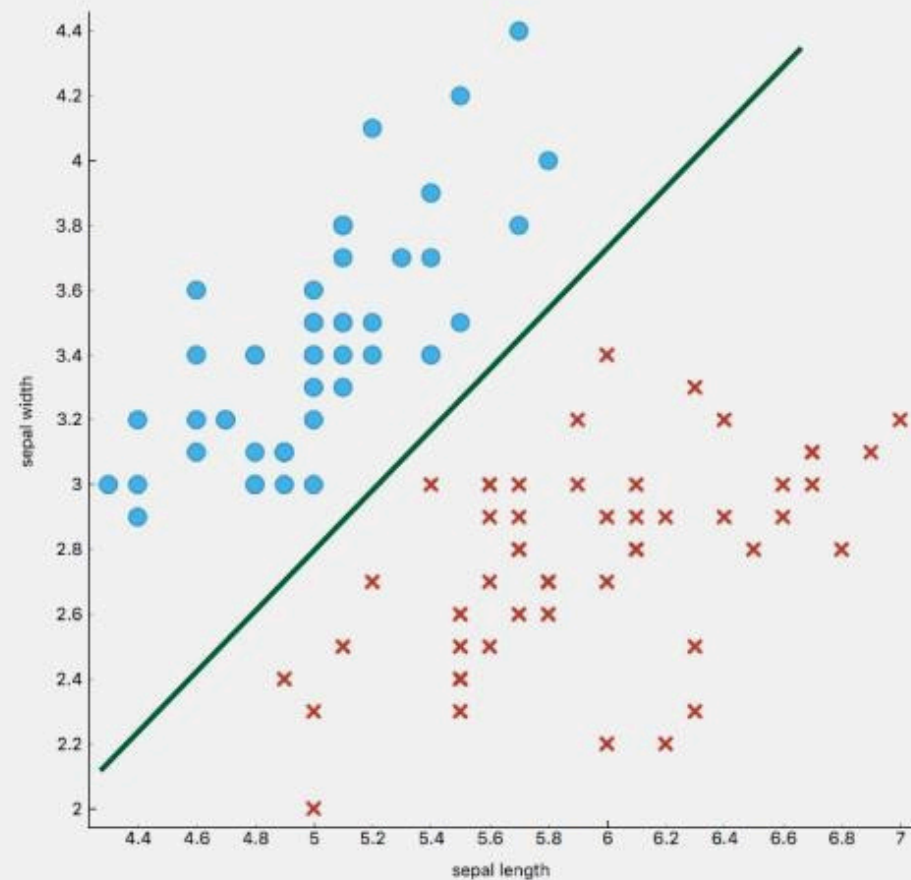
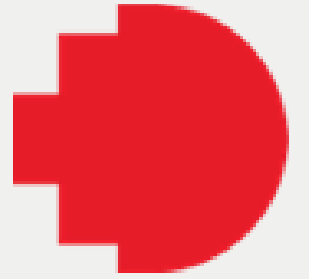
Ockham's Razor



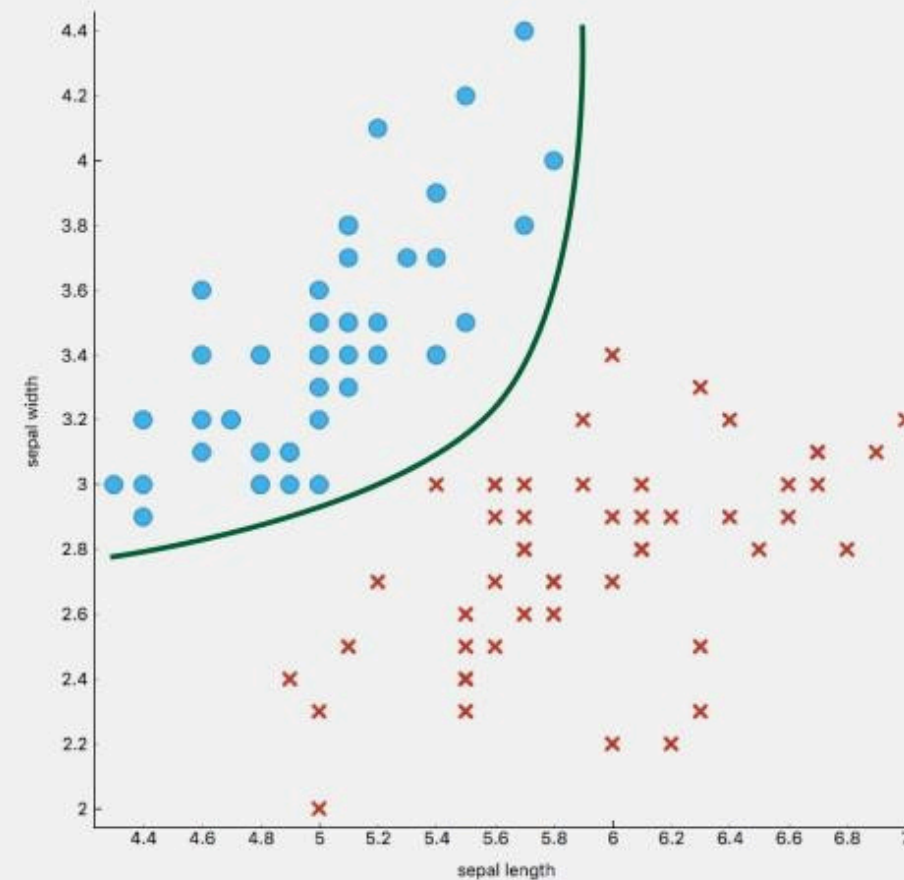
- The **principle of Ockham's Razor** is to prefer the **simplest hypothesis** that is ("reasonably") consistent with the experience



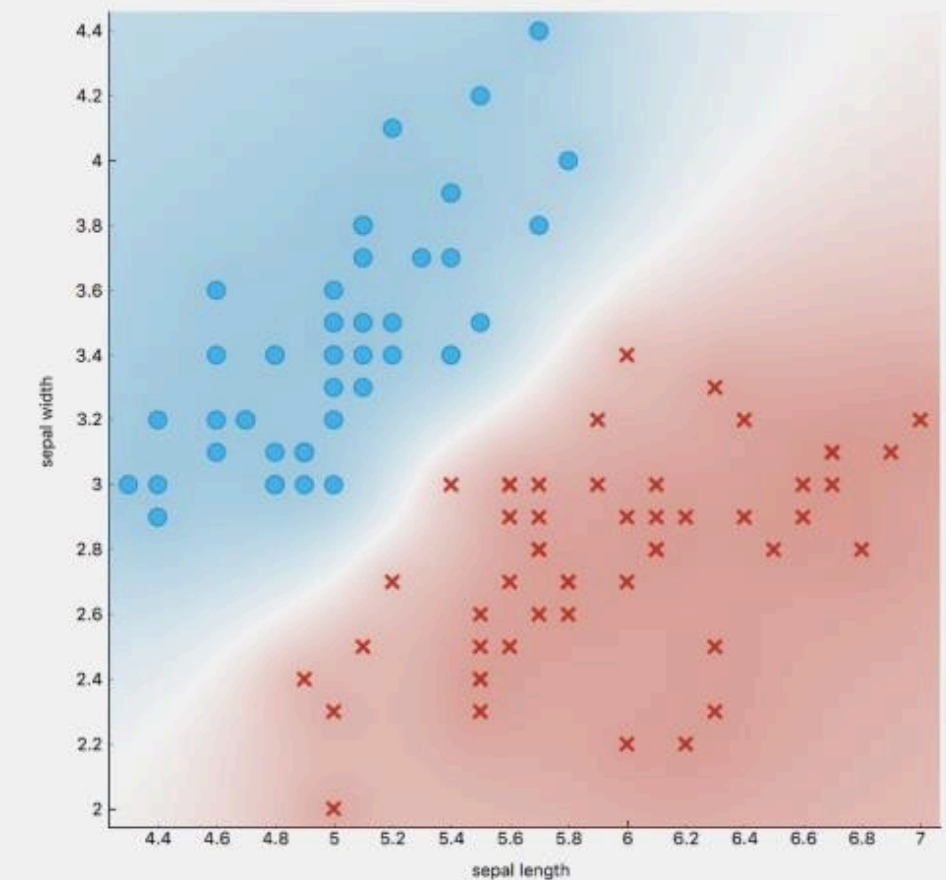
Ockham's Razor



Hypothesis 1



Hypothesis 2



Hypothesis 3

Given that all three hypotheses has zero training error, Ockham's razor says that we should choose the "simplest" hypothesis. We will cover how to measure "simplest" later in the course.

True error & generalization

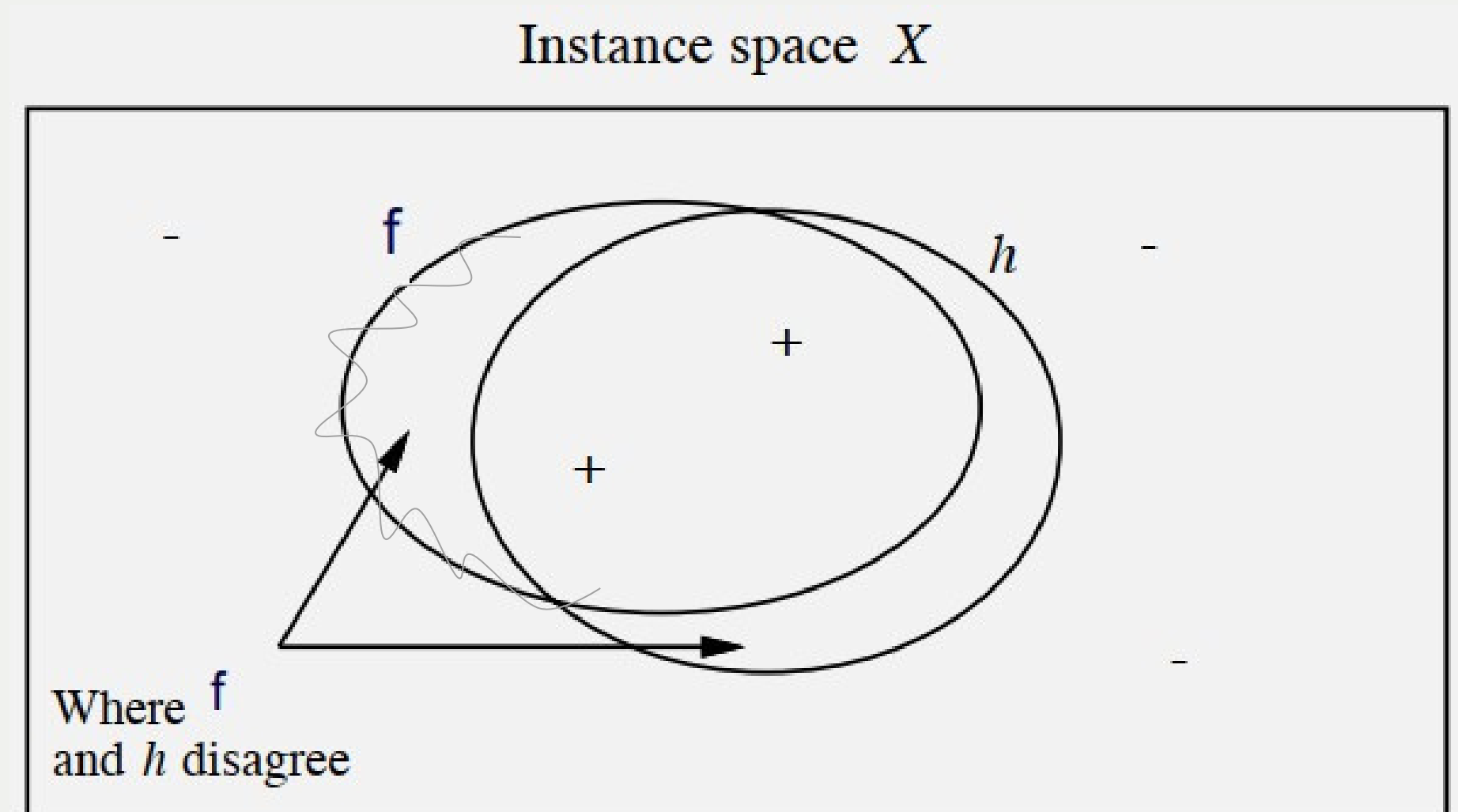
Can the model be “effectively” used to predict for unseen data?

True hypothesis & true error



What is the error of the hypothesis h on the five data points provided?

Is the true-error zero?



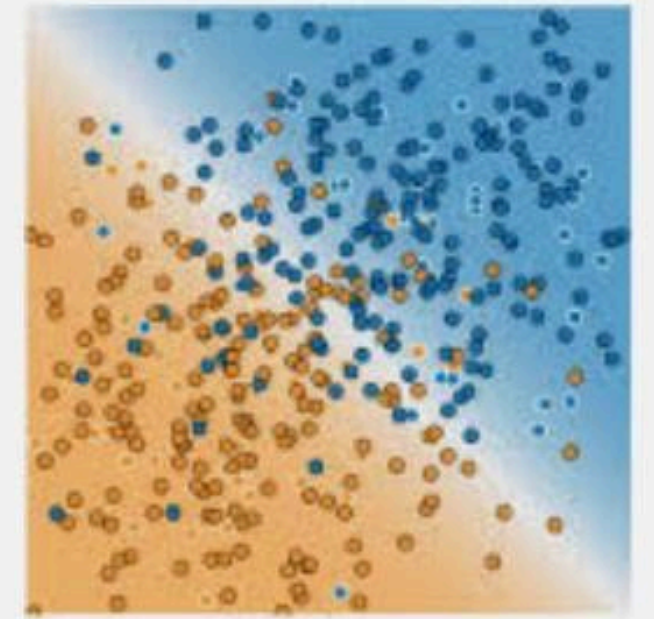
Generalization

ML algorithm must perform well on new, previously unseen inputs (not just those on which our model was trained): **generalization**.

- Generalization error is related to the **true error** of a hypothesis (cannot be measured).

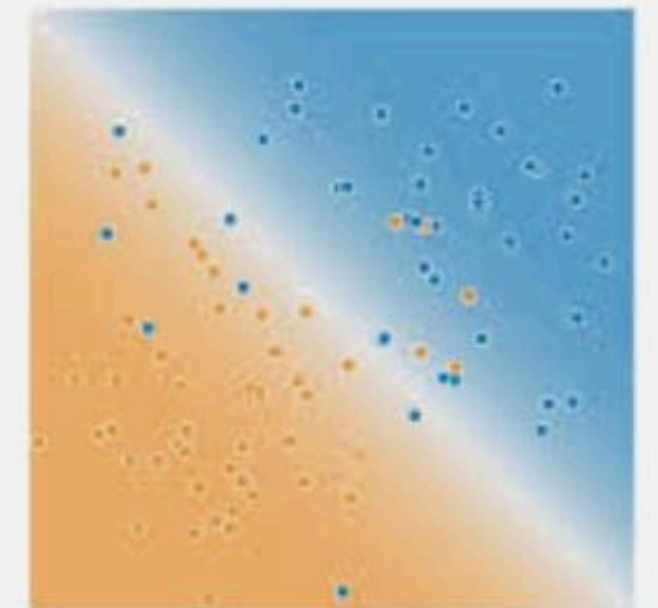
$$h^*(\mathbf{x}) \approx f(\mathbf{x})$$

- Estimate its **performance on a test set** collected separately from the training set.



Training Data

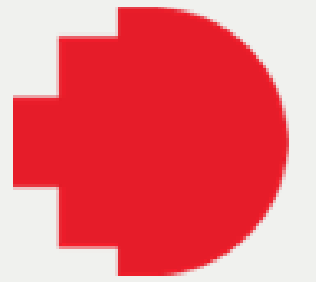
Select the hypothesis with train data



Test Data

Measure performance with test data

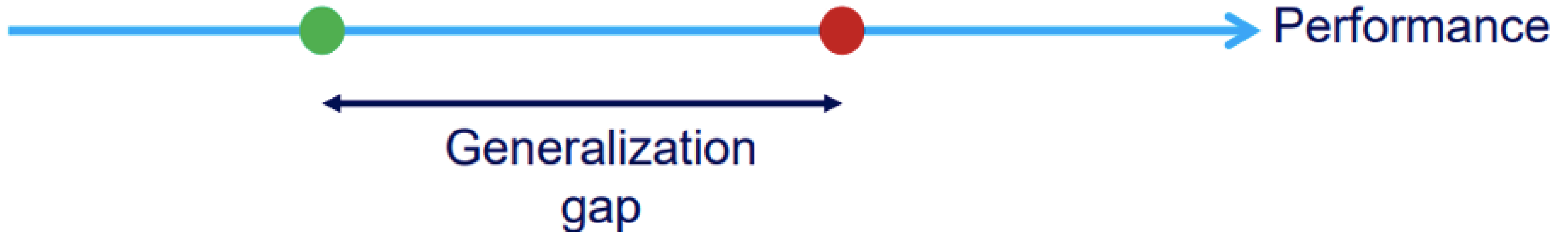
Performance of an ML algorithm



- Make the training error small
- Make the gap between training and test error small (generalization)
- Model capacity: under- or over-fitting.

Test set
performance

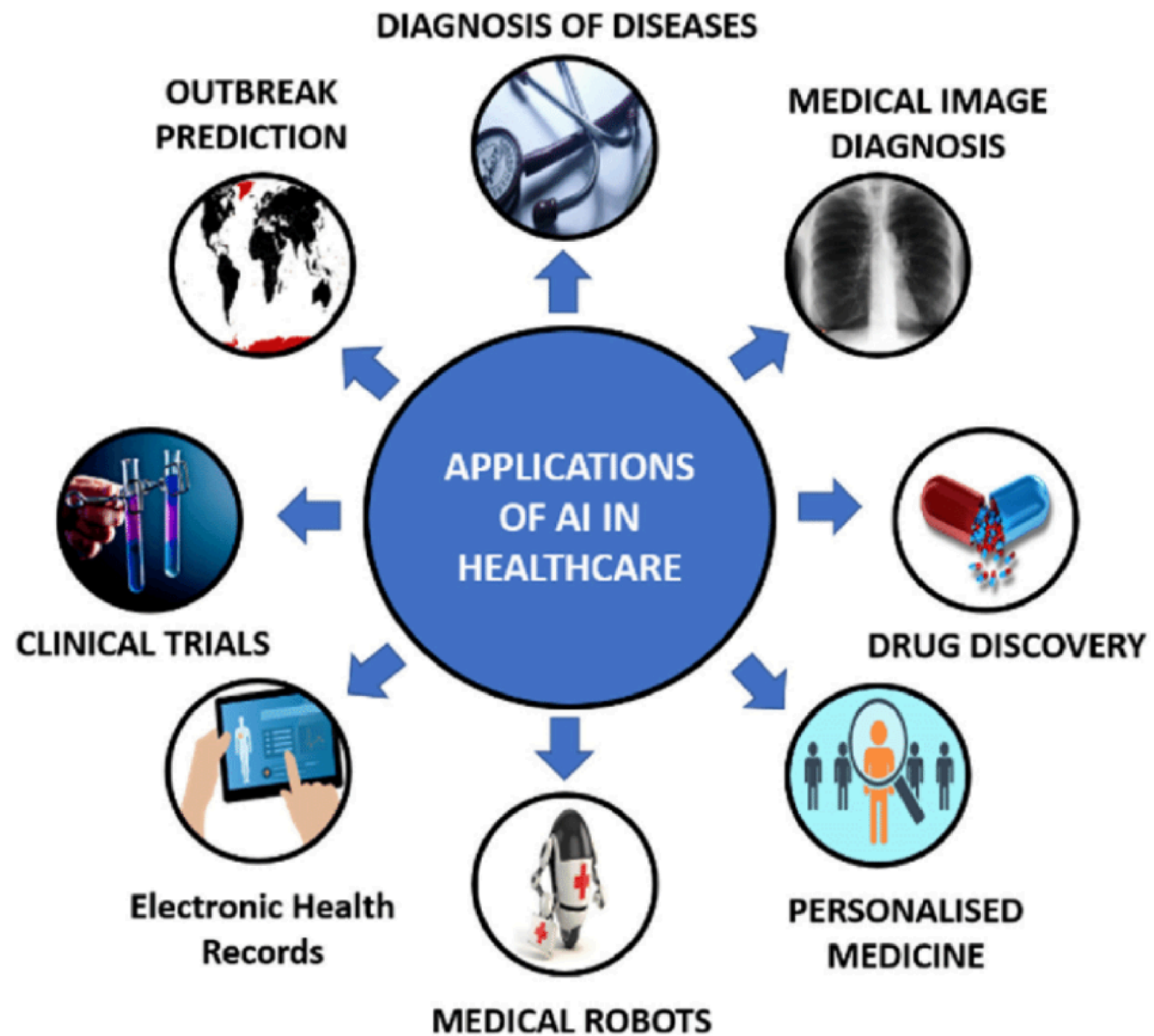
Train set
performance



What is ML type?

Please join by scanning the QR code



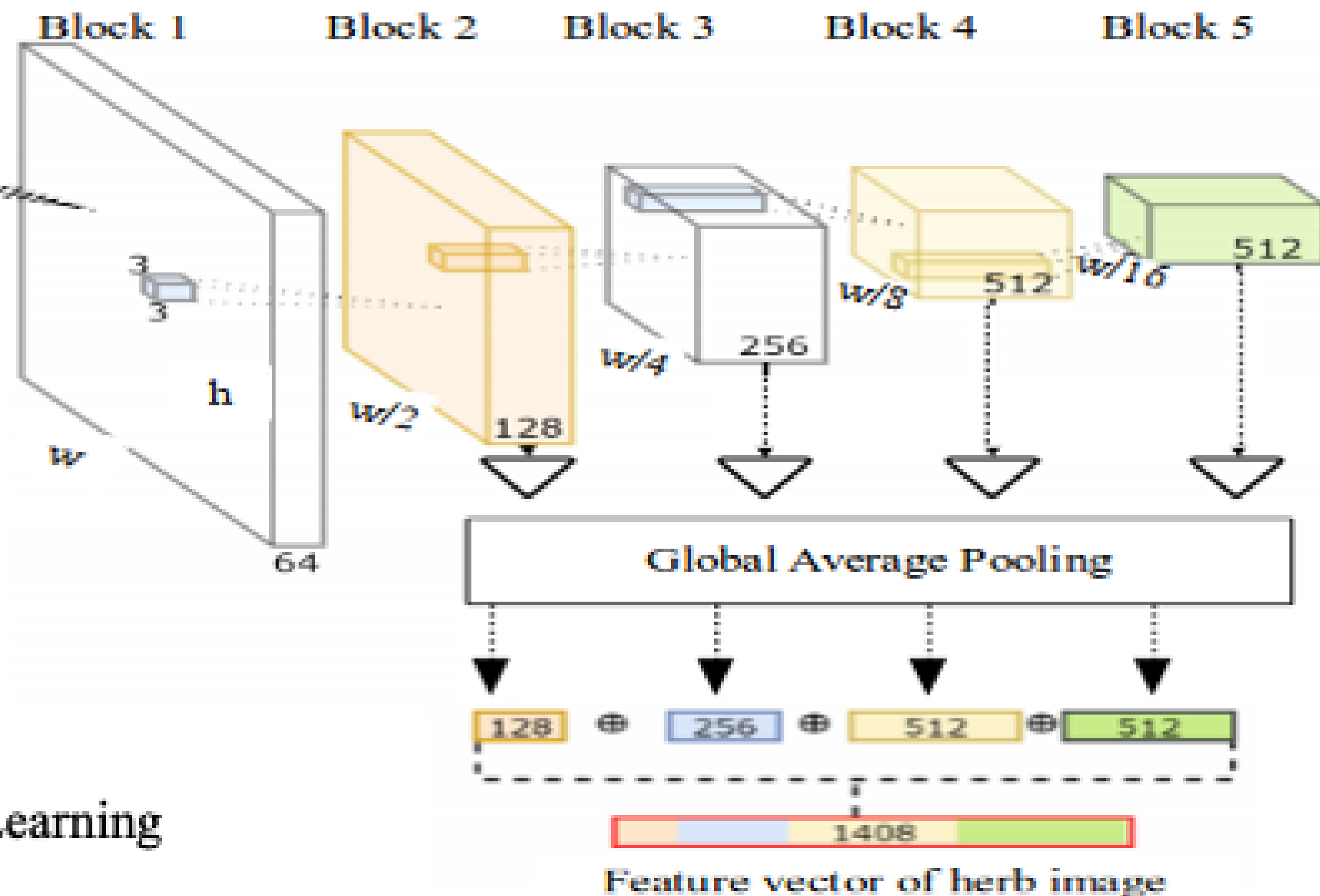
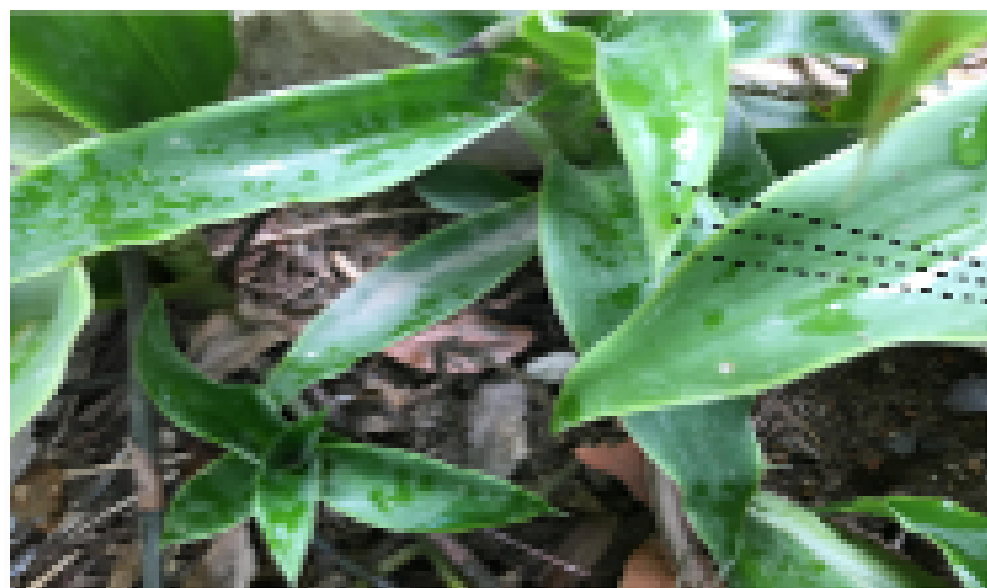


Deep Convolutional Feature in Vietnamese Herbal Plant Recognition

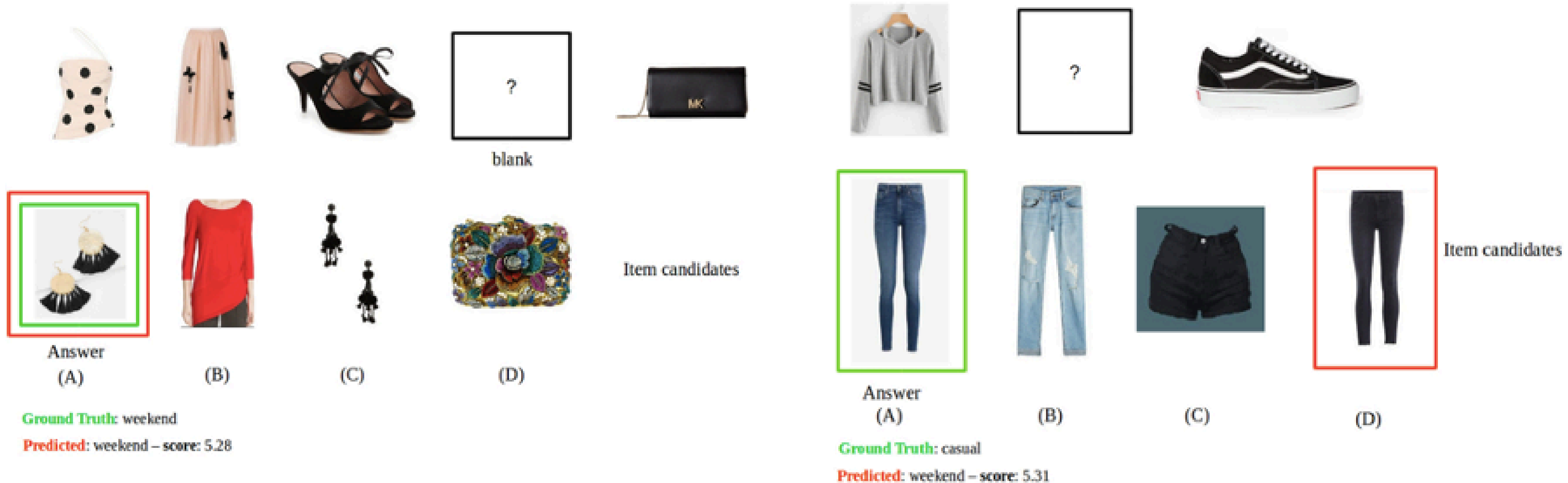
Hoang-Anh Vo

Hoa-T Dang

Thien-Bao Nguyen



Outfit Compatibility prediction



(a) Some examples of the correct cases obtained by our method

(b) Some example of the failed cases obtained by our method

Improtant notes

Ultimate judgement



The core challenge of ML is **NOT ONLY**:

- Collecting data
- Running algorithms
- Maximizing performance

Ultimate judgement



The core challenge is in:

- Deciding what data to use
- Deciding if an algorithm is suitable
- Deciding the most suitable performance measure
- Deciding which hypothesis is “the best” to use for a task
- Making an **ultimate judgement** of how to approximate the unknown target function

Ultimate judgement



The core challenge is in **analysis** and **evaluation**.

- This is the focus of the course
- Running algorithms is necessary and important, but not the top priority

This balance is best thought of as:

- We are not looking for the “best hypothesis”
- We are looking for the “best hypothesis for the task that you can **justify**”

Summary



Introduced Machine Learning

- Definition
- Applications
- Types of ML (supervised, unsupervised, semi-supervised)

Revision



- What are the key ingredients of a general ML recipe?
- What is the difference between hypothesis space and optimal hypothesis?
- What are the main types of ML?

Next week

- Regression

Thank you!